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Infrastructure-Based LiDAR Monitoring for Assessing Automated Driving Safety

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Abstract

The successful deployment of automated vehicles (AVs) has recently coincided with the use of off-board sensors for assessments of operational safety. Many intersections and roadways have monocular cameras used primarily for traffic monitoring; however, monocular cameras may not be sufficient to allow for useful AV operational safety assessments to be made in all operational design domains (ODDs) such as low ambient light and inclement weather conditions. Additional sensor modalities such as Light Detecting and Ranging (LiDAR) sensors allow for a wider range of scenarios to be accommodated and may also provide improved measurements of the Operational Safety Assessment (OSA) metrics previously introduced by the Institute of Automated Mobility (IAM). Building on earlier work from the IAM in creating an infrastructure- based sensor system to evaluate OSA metrics in real- world scenarios, this paper presents an approach for real-time localization and velocity estimation for AVs using a network of LiDAR sensors. The LiDAR data are captured by a network of three Luminar LiDAR sensors at an intersection in Anthem, AZ, while camera data are collected from the same intersection. Using the collected LiDAR data, the proposed method uses a distance-based clustering algorithm to detect 3D bounding boxes for each vehicle passing through the intersection. Subsequently, the positions and velocities of each detected bounding box are tracked over time using a combination of two filters. The accuracy of both the localization and velocity estimation using LiDAR is assessed by comparing the LiDAR estimated state vectors against the differential GPS position and velocity measurements from a test vehicle passing through the intersection, as well as against a camera-based algorithm applied on drone video footage It is shown that the proposed method, taking advantage of simultaneous data capture from multiple LiDAR sensors, offers great potential for fast, accurate operational safety assessment of AV's with an average localization error of only 10 cm observed between LiDAR and real-time differential GPS position data, when tracking a vehicle over 170 meters of roadway.

Introduction

As automated vehicles (AVs) are deployed on public roads, some infrastructure owner-operators (IOOs) may wish to deploy equipment in parallel that can provide an off-board, infrastructure-based assessment of AV operational safety. This is especially true in areas with heavy traffic such as intersections, where accidents involving human-driven vehicles are common, and areas with high concentrations of vulnerable road users (VRUs). The Institute of Automated Mobility (IAM) is developing an operational safety assessment (OSA) methodology, which included defining a comprehensive set of OSA metrics [1] and evaluating the safety envelope-type metrics using simulation methods verified with real-world data [2]. The latter data were obtained from monocular cameras installed at an IAM testbed intersection in Anthem, AZ, but additional sensor modalities have been tested to evaluate future sensor suites that could be installed across AZ [3]. The operational safety assessment requires an accurate measurement of the position, velocity, and acceleration of the AVs.

In recent times, Light Detection and Ranging (LiDAR) sensors have become of great interest to many object detection and tracking projects. LiDAR is a time-of-flight sensing technology that pulses low-power, eye-safe lasers, with

FIGURE 1 Intersection in Anthem, Arizona, used as the testbed for experiments using LiDAR infrastructure in this paper.

sensors measuring the distance and position of various objects using reflected intensity as well as round trip time. 3-D point cloud images are generated by the sensor, which can then be analyzed with computer perception software to facilitate real-time, data-driven decision-making. With a reported maximum error of 10 mm relative to each sensor, coupled with high sampling frequency of nearly 1.5 million points per second [3], LiDAR provides a robust and precise framework for remote sensing and mapping applications. The IAM incorporated a temporary installation of LiDAR technology in its testbeds to help assess the potential for improvement of measurement uncertainty of vehicle position measurements. In particular, the current LiDAR testbed con sists of three Luminar H3-Series Prototype LiDAR sensors mounted at different corners of the intersection between Gavilan Peak Pkwy and Daisy Mountain Dr in the city of Anthem, Arizona. The H3 sensors chosen for this project are dynamically configurable dual scan sensors that can detect objects farther than 200 meters at less than 10 % reflectivity The intersection area used in this paper is as shown in Figure 1 below.

This paper presents a method for processing LiDAR point cloud data taken at the intersection, to obtain velocity and position estimates for vehicles traveling across the intersection over an extended period. Specifically, state information for each vehicle in the intersection is obtained using an Interacting Multiple Model Joint Probabilistic Data Association (IMM-JPDA) filter [4, 5] applied to each point cloud image from a packet capture stream, as described subsequently in the paper. The positions and velocities of target vehicles are estimated using the filter on all three sensors (synchronized in time), and these measurements are compared with two sources: (1) measurements from an RT differential GPS unit mounted in the center of the subject vehicle, and (2) measurements from an artificial intelligence and computer vision-based algorithm using video from cameras mounted aboard a drone. The RT differential GPS unit installed in the test vehicle utilizes precise point positioning (PPP) to achieve position accuracy within 10 cm and camera video tracking on the corresponding vehicles to evaluate the accuracy of LiDAR sensor information.

Literature Review and Problem Description

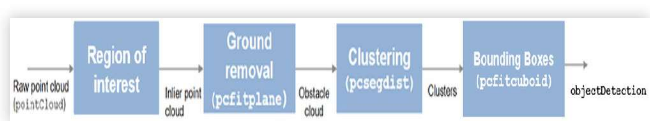
Introduction - Applications of LiDAR Infrastructure in Modern AV Systems

As described in the previous section, LiDAR is a technology that uses time-of-flight readings from electromagnetic pulses (traveling at the speed of light) to measure distances of any detected objects over multiple orientations. Published results describe implementation of LiDAR infrastructure for improving sign visibility in foggy areas [6, 7], which do not depend on ambient light since LiDAR sensors generate their own energy. These results show that in traffic infrastructure, LiDAR sensors are robust to disturbances (primarily from weather effects) and can detect and associate moving objects across multiple point clouds. More recently, LiDAR sensors have also been deployed in commercial AV's, such as the Waymo fleet (using the Honeycomb LiDAR [8]). In many LiDAR testbeds, such as the ones described above, the working of data processing algorithms can be summarized in three major steps: (1) segmentation, (2) clustering, and (3) data association/tracking. Figure 2 below shows these steps as implemented on MATLAB for vehicle detection.

LiDAR has also become a popular choice for use in traffic intersections, with some recent papers focusing on the efficiency of packet transfer within a network of LiDAR sensors [9] while others focus on aspects relating to the accuracy of detections generated clustering algorithm when LiDAR data are provided in real time while distinguishing between different road features such as vegetation and debris [10, 11]. These papers have established that LiDAR infrastructure is a viable and efficient choice for traffic safety evaluation; however, a key aspect that remains to be addressed is the experimental validation of LiDAR estimated position and velocity information when compared with ground truth data, particularly when considering large intersections with heavy traffic.

In this work, we propose a LiDAR bounding box detection method that is validated by comparing estimated position and velocity values with real time GPS data, for a vehicle moving through a crowded intersection. It is proposed that the LiDAR estimated data are accurate enough to assist vehicles in achieving operational safety metrics such as the Minimum Safe Distance Violation (MSDV) metric and the Time Headway Violation (THWV) metric. From the prior work done by the IAM to establish a driving safety infrastructure [1], an MSE

FIGURE 2 Bounding box detection model for LiDAR point cloud data.



and THWV occurs according to equations as defined in Table 2 of the paper by Elli, Wishart et al. [2].

A LiDAR point cloud can be used to obtain the location coordinates of each vehicle in the X, Y and Z axes which can be used directly to estimate a time headway violation, while an MSDV can be calculated using an accurate localization algorithm to estimate d_{min}^{long} and d_{min}^{lat} within a few tenths of a meter (based on chosen values for ρ_1, ρ_2 and μ, as defined in the IAM's previous work [1, 2]).

Methodology - Description of LiDAR Processing Techniques

Segmentation From a 3D point cloud, the objective of segmentation is to attribute sets of points to predefined candidate object classes. This is particularly useful in separating ground points from non-ground points in each scene. As illustrated subsequently in this paper, the removal of ground points is an important pre-processing step to identify and detect obstacles, especially in traffic intersections where vehicles share the same roadway, but move in different directions and velocities. There have been multiple research results describing ground plane segmentation methods; this includes using an elevation map created by observing the relative height difference between points, as proposed by researchers at Stanford University [12], as well as more recent attempts to apply iterative segmentation from an initial set of known ground points [13, 14]. One example of the latter is the simple morphological filter (SMRF) method, developed at the University of California, Santa Barbara. The SMRF algorithm can be summarized as follows

- Creation of a minimum elevation surface, by finding the lowest elevation value for each pixel (or X-Y point).
- Iterative application of a morphological opening on the minimum surface map, (using a predefined disk- shaped structural element) to create a binary mask function (separating ground from nonground based on an elevation threshold value)
- Application of the binary mask on original point cloud, to create a provisional ground surface on the point cloud.

Clustering Once a given point cloud has been separated into ground and nonground points, it may then be processed to identify features and objects based on point density. This is achieved through applying a Euclidean clustering algorithm to the point cloud. In this example, the clustering algorithm partitions the non-ground points into subgroups, or clusters, based on a minimum segmentation distance. Each point in the point cloud is given a cluster label, classifying it as part of larger objects by creating a bounding box around points sharing the same cluster label (within a predetermined segmentation distance, specified in meters, to separate one cluster from another).

To generate clusters, batches of raw points from the ground-segmented point cloud are taken one at a time to build a K-Dimensional tree,, which is a data structure comprising of multiple K-dimensional points (in this case, 3-D points).

FIGURE 3 Cluster detection model.

```
clusters = ∅
for p ∈ P do
    Q = ∅
    if p.status ≠ processed then
        Q.add(p)
        for q ∈ Q do
            q.status = 'processed'
            N = GetNeighbors(q, ε)
            Q.addAll(N)
        if size(Q) ≥ minPts then clusters.add(Q)
```

The K-Dimensional (or K-D) tree is divided into non-overlapping regions such that each point may be classified as being part of a given region. It is also a quick way of organizing data points to accelerate detection of nearest neighbors for each point, based on the region to which it belongs in the KD-tree. Using this structure, the cluster detection algorithm can be summarized as seen in Figure 3 [15].

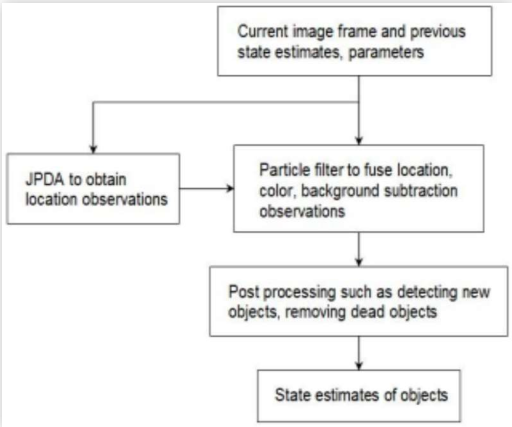
Where p represents a single point out of P points within the point cloud, q represents an unprocessed point within batch Q, N represents the neighboring points within clustering distance ε, and *minPts* is the minimum number of points required within ε meters of one another to be classified as an object cluster. This clustering algorithm is borrowed from the Point Cloud Library (PCL), an open-source standalone project [16, 17].

Data Association and Tracking With an established methodology for creating clusters in each point cloud, the objective of data association is to correlate clusters from successive point cloud images (within a large packet capture) to establish a track list of objects moving between frames. This is an especially important step in estimating the velocities of moving objects. Estimating the velocities of moving objects is achieved by using the difference in timestamp (between two frames) along with the calculated shift in position. Typically, tracking a bounding box across multiple frames is achieved using a combination of Kalman filters.

In this work, we consider the use of an Interacting Multiple Model (IMM) filter, a powerful framework making use of the Kalman filter to track the motion and shape of various subjects within a given image. While this method promises a fast and stable tracking response, proven to perform better than any single model algorithm in complex tracking problems [18], a major limitation of this algorithm is that it tends to produce erroneous outputs in the presence of clutter, which occurs when multiple tracking subjects are located extremely close to one another. This limitation greatly affects the performance of target tracking within an intersection, where multiple vehicles are situated very close to one another.

To address this limitation, this work proposes the use of a Joint Probabilistic Data Association (JPDA) filter, together with the IMM filter. The joint probabilistic data association (JPDA) estimates the states of objects based on all possible associations between objects and observations. The working of the algorithm, implemented on MATLAB, is as illustrated in Figure 4 below [19, 20].

FIGURE 4 The JPDA Tracking algorithm.



As shown in Figure 3, the JPDA algorithm compares location information for different tracking subjects between successive frames, using a particle filter on background-subtracted images (in this case, point cloud frames). This particle filter offers greater precision when attempting to differentiate between multiple tracking subjects within proximity.

The present work proposes the use of the IMM and JPDA algorithms together in an IMM-JPDA filter implementation. The JPDA filter acquires detection information from the preliminary IMM filter outputs, assigning a distinct track ID to each bounding box detections associated with previously detected clusters, corresponding to moving vehicles. This implementation is based on a well-established method established in the seminal paper by Dr. Henk A.P.. Blom and Dr. Edwin A. Bloem [21].

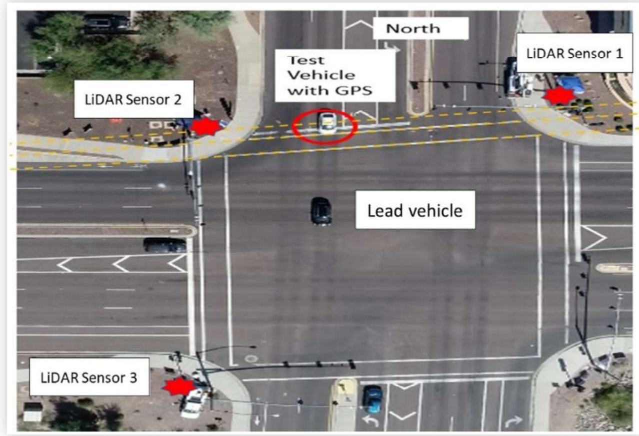
Experimental Testbed and Data Analysis

System Infrastructure and Data Capture

The sensing infrastructure used in these experiments is comprised of three Luminar H3 LiDAR sensors (each mounted at different corners of the intersection), a deployed drone with a video camera, and a test vehicle equipped with mobile sensors to collect ground truth data (with a lead vehicle also designated, travelling right ahead of the test vehicle). Figure 5 below shows an aerial view of the intersection with the positions of LiDAR sensors and test vehicle marked.

Each of the Luminar LiDAR units shown in Figure 4 covers 30 degrees in elevation and 120 degrees in azimuth and is connected by Ethernet to synchronize detection data. Data from these sensors are returned in the form of a packet capture (PCAP) file, which in turn can be separated into individual frame captures using Wireshark or any network packet capture analysis tool. The LiDAR sensors used were provided by Luminar, with a total range of nearly 220 m.

FIGURE 5 Data capture setup at intersection (drone image).



For video footage, a drone was utilized to hover over the study intersection to capture video footage of test vehicles and other traffic passing through the intersection by using a ground facing camera (as seen in Figure 5, which was taken by the drone).

Finally, the RT differential GPS unit was mounted on a test vehicle, circled in Figure 5, to provide accurate ground truth data (down-sampled to 10 Hz from 100 Hz), as a basis for comparison with the position and velocity information processed using drone video processing and LiDAR captures respectively. Since LiDAR readings are returned in the form of range and azimuth/elevation angle readings (within its local frame), pose information provided by Luminar for each sensor is used to transform these readings into global coordinate values, using a rotation matrix.

Data Processing and Analysis

Once LiDAR data have been obtained in the form of a packet capture (PCAP) file, data may be observed in the form of point clouds representing each frame of the capture. Figure 6 shows a sample point cloud image as seen on Luminar EnVision software, containing the test and lead vehicles as observed on Figure 5 (with sensor 1 as the origin)

To facilitate the comparison with test vehicle GPS data, the LiDAR point clouds are also sampled at 10 Hz. Once provided to MATLAB, a native Euclidean clustering algorithm is applied to the data in order to generate clusters, which in turn are used to create bounding boxes. In the present implementation, it is desired that these bounding boxes will be used to identify clusters that are classified as vehicles. This classification is performed using a pretrained vehicle detector trained using the PointPillar neural network (as provided by MATLAB) [10]. Figure 7 shows the clustering as performed on the lead and test vehicle, on one frame of data.

Following the clustering step, An IMM filter is then initialized using a constant velocity and constant turn-rate model. This filter is initialized together with a JPDA tracker, which takes the outputs of the IMM filter models, and attempts to update and correct the probabilities within each model based on the confidence of object detections associated with

FIGURE 6 Point Cloud data from sensor 1 (lead and test vehicle).

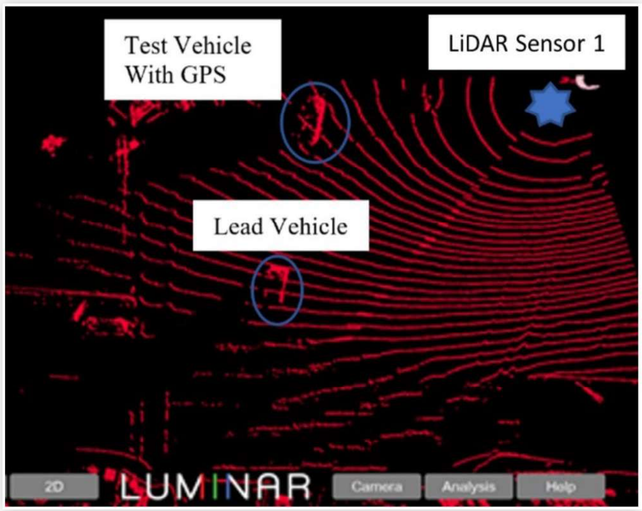
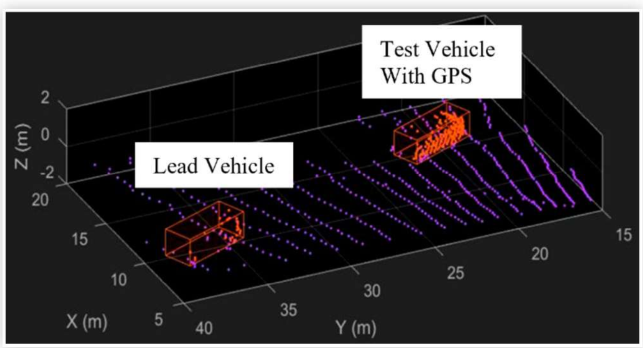


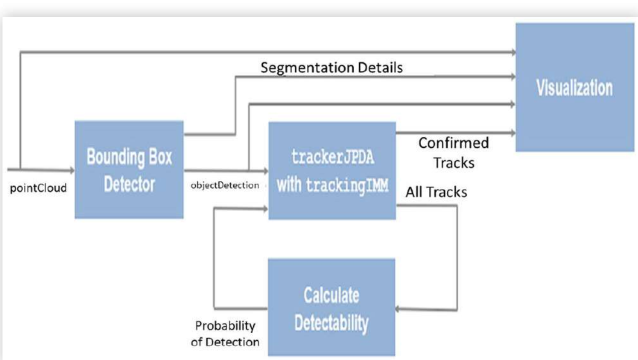
FIGURE 7 Vehicle clustering from point cloud data, sensor 1 (lead and test vehicle).



previous frame data. The working of the algorithm is summarized in Figure 8.

Thus, using this framework, a list of confirmed "tracks" (i.e vehicle bounding boxes detected over multiple successive frames) is generated, with each track containing state estimations for position and velocity of the associated object (calculated using the unscented Kalman filters in the IMM tracker). These state estimates are documented for the test vehicle, as well as a few others passing through the intersection in order to compare accuracy with vehicle GPS data as well as estimated position and velocity data from the drone video capture.

FIGURE 8 IMM/JPDA tracker set up (from MATLAB).



Main Results and Discussion

With the sensing infrastructure set up as described in the previous section, LiDAR data are captured at the intersection and provided to MATLAB in the form of point cloud frames.

These frames are processed one after another by the IMM-JPDA tracker, which generate estimated position and velocity values of bounding box clusters, associated with vehicle detections.

Data are first analyzed for the test vehicle, for approximately 5 seconds of time during which both the lead vehicle and test vehicle pass through the intersection, as seen in Figures 5 and 6. Firstly, using the SMRF filter, ground plane data are subtracted from each point cloud. The Euclidean clustering algorithm is then applied to the segmented point cloud, to visualize and fit bounding boxes for each vehicle in the point cloud. A sample frame from the segmented point cloud is shown in Figure 9.

The IMM-JPDA filter is then applied, identifying bounding box detections that appear associated with one another across multiple frames. With a sampling rate of 10 Hz, there are a total of 50 frames covering the 5 second period; of these 50 frames, vehicle track information for the test vehicle is captured accurately for around 30 frames. Position and velocity data reported by the LiDAR tracker are recorded for these frames and are compared with the observed ground GPS data collected onboard the vehicle.

Figure 10 below shows the comparison between position data reported by the GPS and LiDAR tracker respectively. In this case, the origin (0 m) is set as the point at which the test vehicle enters the intersection, with distance travelled by the vehicle measured from this point.

As seen in Figure 10, over the 30 frames (~3 seconds) of compared data, position data measured by the LiDAR follows GPS position data very closely, with a maximum localization error of 0.5 m.

FIGURE 9 Background subtracted LiDAR image for test and lead vehicle (ground points subtracted from Figure 6).

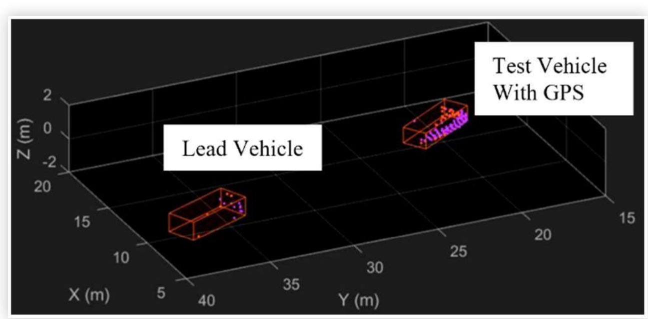
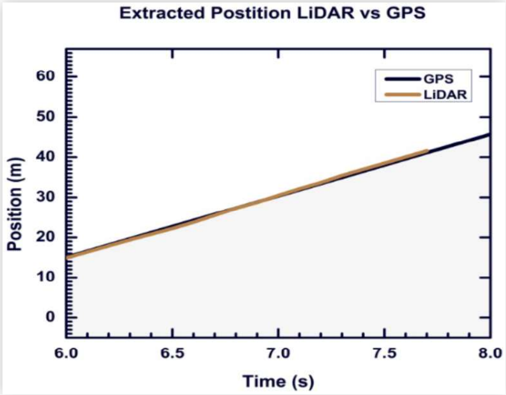


FIGURE 10 Position data comparison between LiDAR and GPS data.



Similarly, Figure 11 shows the comparison between GPS velocity and the velocity estimated by the LiDAR tracker, shown over 20 frames of data (2 seconds). Estimated velocity from drone video data are also shown in the figure.

As seen in Figure 11, the velocity estimated by the LiDAR almost accurately follows the corresponding GPS velocity values, with a maximum error of 1m/s near the beginning of the video (for about 0.2 seconds). The drone video readings are also similar over this period, although the LiDAR tracking results appear to follow the GPS data more closely. Figure 12 plots the maximum error between LiDAR and GPS data.

As seen in Figure 12, localization errors are observed to be at a maximum of 0.5 m. The average localization error is found to be around 0.1 m (or 10 cm).

To further examine the accuracy of the proposed tracking method (specifically, over a longer time and distance) a track is also performed on a preceding vehicle passing through the intersection, traveling in the same direction as the test vehicle. Figure 13 shows a frame from the drone video capture highlighting this preceding vehicle, which is assigned track #60 in this image. (With lead and test vehicles also visible as tracks 69 and 83 respectively).

FIGURE 11 Velocity data comparison between LiDAR, GPS, and overhead/eagle eye camera (for test vehicle).

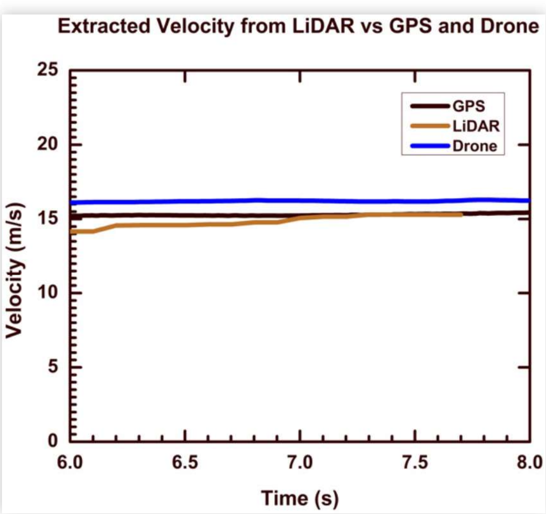


FIGURE 12 Plot of error between LiDAR and GPS data.

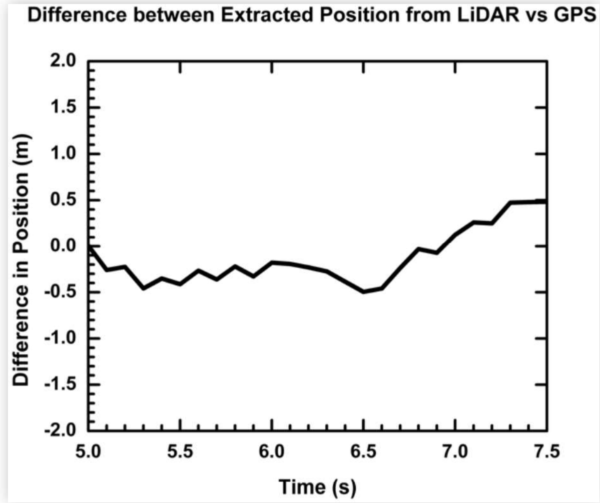
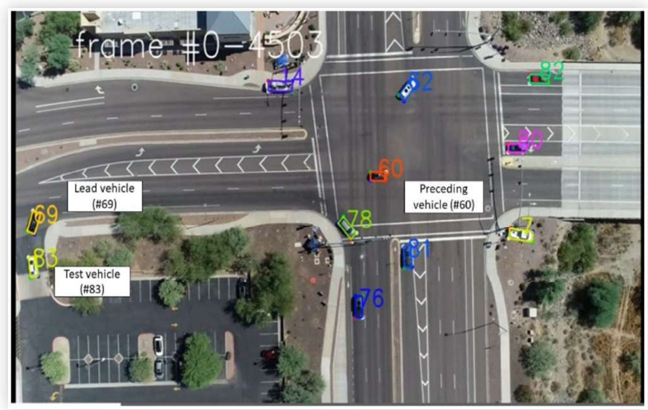


FIGURE 13 Drone image of preceding vehicle, labeled #60 in the intersection, that was by LiDAR to evaluate detection range.

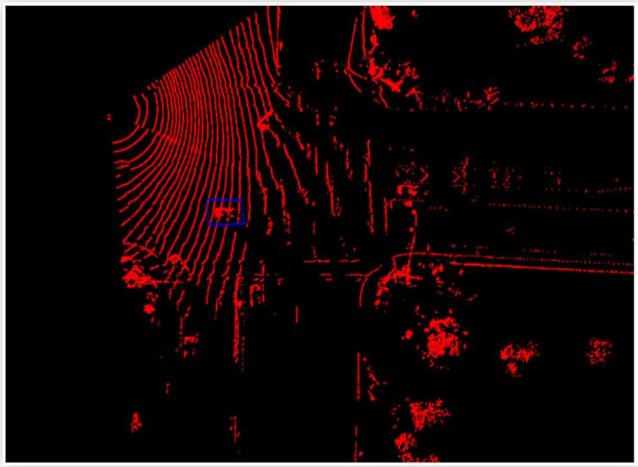


In this case, LiDAR tracking information is compared with position and velocity estimates from the drone video data within the intersection. Velocity and position estimates are obtained for each vehicle using a deep neural network to estimate the vehicle motion and associate detected object instances (in the form of 3D bounding boxes) across adjacent frames. From these bounding boxes, location and speed, in the world coordinate system, are obtained by applying a Kalman filter with rigid body kinematics to smooth the vehicle states. All video processing was performed using computer vision techniques, by the IAM team [1].

A frame of LiDAR data with the preceding vehicle (corresponding to Figure 13 from the drone video data) is shown in Figure 14, with the preceding vehicle in a blue box.

Position data for the preceding vehicle is obtained from the LiDAR tracker for a total of 120 frames (12 seconds), over nearly 160 meters (with 0 m point taken as the pedestrian line on the left side of the intersection in Figure 13), with the LiDAR to vehicle distance being over 170 m when LiDAR data acquisition of this isolated vehicle stopped. This demonstrated this LiDAR's capability to track vehicle over a significant distance,

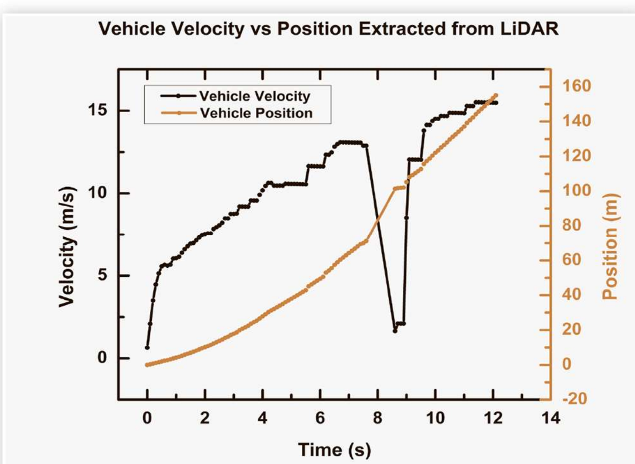
FIGURE 14 Background subtracted LiDAR image from single (preceding) vehicle track.



which would benefit the tracking and monitoring of traffic for operational safety purposes. The LiDAR can consistently detect vehicles and objects over 220 m in range during the data acquisition. A plot of the extracted position data and corresponding velocity estimated by the LiDAR tracker is shown in Figure 15.

As seen in Figure 15, distance and velocity show a steady increase, although it is to be noted that there a significant drop around 8-9 seconds; this is because the LiDAR tracker must initialize a new track for the vehicle during this time once the preceding vehicle has crossed the intersection, and it takes around 0.5 seconds for this track to stabilize. The position and velocity data are found to be fairly accurate when compared to the estimated position and velocity from drone video data when measured in the intersection (i.e., until 8 seconds), and the LiDAR tracker can detect the vehicle beyond the intersection over a total distance of 160 meters, which is the total distance the preceding vehicle (marked in Figure 14) travels within the LiDAR sensor's frame of view.

FIGURE 15 Position and velocity data extracted from LiDAR data form preceding vehicle, marked in Figures 13 and 14. LiDAR data acquisition for this isolated vehicle stopped at 160 m, while the vehicle detection range is over 220 m.



Conclusion

In this paper, we present a method for using a LiDAR sensing infrastructure to accurately detect, cluster and track vehicles passing through an intersection with speeds up to 33 miles per hour (or 15 meters per second). Data captured at the intersection is preprocessed using MATLAB, specifically to segment non-ground data, and following this step, a simple Euclidean clustering algorithm is applied to this data to generate bounding boxes corresponding to each cluster identified as a vehicle (using a pretrained PointPillar network). This clustering is performed using a nearest neighbor querying algorithm (described earlier in the paper) which is done after organizing segmented point cloud data in a KD-tree structure[10] Position and velocity values for target vehicle are estimated by an IMM-JPDA tracker using the clustered point cloud data as input, found to be sufficiently accurate when compared to corresponding position and velocity values estimated by the GPS data (measured directly on the test vehicle) as well as processed drone video data. Specifically, accurate position measurements provide a reliable basis for efficient operational safety metric evaluation in traffic, such as the THWV (which may be directly calculated from position and velocity measurements) and the MSDV metric (which can be estimated accurately within a few tenths of a meter subject to road conditions and assumptions regarding driver behavior, as outlined previously)

As seen in Figure 15, the LiDAR tracker is also shown to be capable of following a vehicle over a range of nearly 160 m. However, in doing so, it briefly loses visibility of the vehicle when it leaves the intersection (at 8-9 seconds in Figure 15) which means that a new track must be initialized once the vehicle is detected again, causing a temporary loss in precision. A potential solution is to incorporate track memory into the processing algorithm, to predict vehicle trajectory during periods where the LiDAR sensors cannot see the vehicle (although this would potentially increase computational overhead and system delays). It is also to be noted that in this work, sensor processing is done individually. The infrastructure at the intersection consists of three LiDAR sensors (primarily to cover blind spots from other sensors) although the processing is done on each sensor one at a time, using the same method. This paper seeks to establish and analyze the effectiveness of the tracking method, rather than the conflation of data from different sources in real time.

It is hoped that future research may build on the framework established in this paper by attempting alternative, more complex clustering and segmentation methods (such as panoptic segmentation, a hybrid method with an existing semantic segmentation network to extract semantic information and a traditional LiDAR point cloud cluster algorithm to split each instance object). Additionally, it is hoped that this method may be applied to evaluation of multiple operational safety metrics (apart from MSDV) in larger intersections, as well as in neighborhoods and residential areas with more pedestrian traffic.

References

1. Altekar, N., Como, S., Lu, D., Wishart, J. et al., "Infrastructure-Based Sensor Data Capture Systems for Measurement of Operational Safety Assessment (OSA) Metrics," *SAE Int. J. Adv. & Curr. Prac. in Mobility* 3, no. 4 (2021): 1933-1944. https://doi.org/10.4271/2021-01-0175.
2. Elli, M., Wishart, J., Como, S., Dhakshinamoorthy, S. et al., "Evaluation of Operational Safety Assessment (OSA) Metrics for Automated Vehicles in Simulation," SAE Technical Paper 2021-01-0868 (2021). https://doi.org/10.4271/2021-01-0868.
3. Guan, H., Li, J., Cao, S., and Yongtao, Y., "Use of Mobile LiDAR in Road Information Inventory: A Review," *International Journal of Image and Data Fusion* 7, no. 3 (2016): 219-242.
4. Bar-Shalom, Y., Willett, P.K., and Tian, X., *Tracking and Data Fusion* (Storrs, CT, USA: YBS Publishing, 2011).
5. Bojilov, L., Alexiev, K., and Konstantinova, P., "An Accelerated IMM-JPDA Algorithm for Tracking Multiple Maneuvering Targets in Clutter," *Information & Security: An International Journal* 9 (2002): 141-153.
6. Wu, S., Wen, C., Luo, H., Chen, Y. et al., "Using Mobile LiDAR Point Clouds for Traffic Sign Detection and Sign Visibility Estimation," *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)* 2015 (2015): 565-568, doi:10.1109/IGARSS.2015.7325826.
7. Kutila, M., Pyykönen, P., Holzhüter, H., Colomb, M. et al., "Automotive LiDAR Performance Verification in Fog and Rain," in *2018 21st International Conference on Intelligent Transportation Systems (ITSC)*, 2018, pp. 1695-1701, doi: 10.1109/ITSC.2018.8569624.
8. Introducing Laser Bear Honeycomb by Waymo, https://waymo.com/lidar/, 2019.
9. Lv, B., Xu, H., Wu, J., Tian, Y. et al., "LiDAR-Enhanced Connected Infrastructures Sensing and Broadcasting High-Resolution Traffic Information Serving Smart Cities," *IEEE Access* 7 (2019): 79895-79907.
10. Zhao, J., Xu, H., Liu, H., Wu, J. et al., "Detection and Tracking of Pedestrians and Vehicles Using Roadside LiDAR Sensors," *Transportation Research Part C: Emerging Technolgies* 100 (2019): 68-87.
11. González-Gómez, K., López-Cuervo Medina, S., and Castro, M., "Assessment of Intersection Conflicts between Riders and Pedestrians Using a GIS-Based Framework and Portable LiDAR," *GIScience & Remote Sensing* 58 (2021): 587-602.
12. Thrun, S., Montemerlo, M., Dahlkamp, H., Stavens, D. et al., "Stanley: The Robot that Won the DARPA Grand Challenge," *Journal of Field Robotics* 23 (2006): 661-692.
13. Zermas, D., Izzat, I., and Papanikolopoulos, N., "Fast Segmentation of 3D Point Clouds: A Paradigm on LiDAR Data for Autonomous Vehicle Applications," in *Proceedings of the 2017 IEEE International Conference on Robotics and Automation (ICRA)*, Singapore, 29 May-3 June 2017, 5067-5073.
14. Pringel, T., Clarke, K., and Mcbride, W., "An Improved Simple Morphological Filter for the Terrain Classification of Airborne LIDAR Data," *ISPRS Journal of Photogrammetry and Remote Sensing* 77, no. 1 (2013): 21-30.
15. Najdataei, H., Nikolakopoulos, Y., Gulisano, V., and Papatriantafilou, M., "Continuous and Parallel LiDAR Point-Cloud Clustering," in *2018 IEEE 38th International Conference on Distributed Computing Systems (ICDCS)*, 2018, pp. 671-684, doi: 10.1109/ICDCS.2018.00071.
16. Rusu, R.B. and Cousins, S., "3D is Here: Point Cloud Library (PCL)," in *2011 IEEE International Conference on Robotics and Automation*.
17. Ramiya, A.M., Nidamanuri, R.R., and Krishnan, R., "Segmentation Based Building Detection Approach from LiDAR Point Cloud," *The Egyptian Journal of Remote Sensing and Space Science* 20, no. 1 (2017): 71-77. https://doi.org/10.1016/j.ejrs.2016.04.001.
18. Wan, M., Li, P. and Li, T., "Tracking Maneuvering Target with Angle-Only Measurements Using IMM Algorithm Based on CKF," in *2010 International Conference on Communications and Mobile Computing*, 2010, pp. 92-96, doi: 10.1109/CMC.2010.239.
19. "Joint Probabilistic Data Association Multi-Object Tracker," https://www.mathworks.com/help/fusion/ref/jointprobabilisticdataassociationmultiobjecttracker.html.
20. MATLAB, *Version R2021a* (Natick, Massachusetts: The MathWorks Inc, 2021).
21. Blom, H.A.P. and Bloem, E.A., "Combining IMM and JPDA for Tracking Multiple Maneuvering Targets in Clutter," in *Proceedings of the Fifth International Conference on Information Fusion. FUSION 2002. (IEEE Cat.No.02EX5997)*, 2002, pp. 705-712 vol. 1, doi: 10.1109/ICIF.2002.1021224.
22. Lang, A.H., Vora, S., Caesar, H., Zhou, L., et al., . "PointPillars: Fast Encoders for Object Detection From Point Clouds," in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 12689-12697. Long Beach, CA, USA: IEEE, 2019.

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